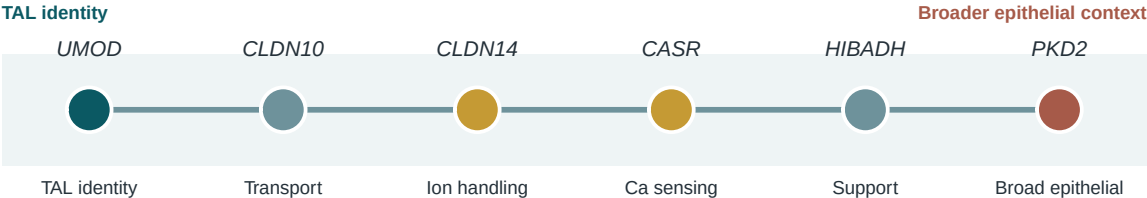


A. P1 candidate role spectrum

Six MAGMA-prioritized genes organized from TAL identity to broader epithelial context



B. P1 expression across audited contexts

Dot size = detected nuclei;
color = mean expression

oop/TAL



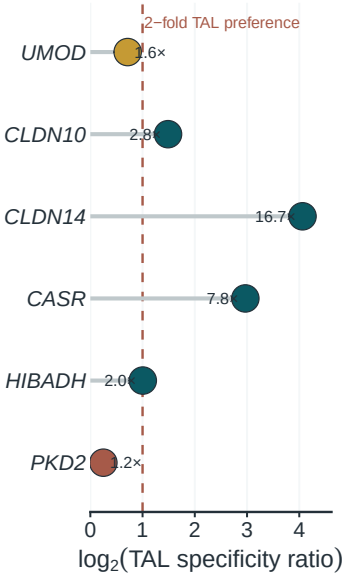
Expression

Low Moderate High
Detected (%)

25 50 75

C. TAL specificity and donor support

Labels = raw TAL specificity ratio
All genes: 3/4 TAL donors



Specificity

Broad Moderate Strong

D. Gene-centric evidence summary

	MAGMA	TAL	Donor	Specificity	Bulk	Role
<i>UMOD</i>	+	Rank 1	3/4	Moderate	no FDR	TAL id.
<i>CLDN10</i>	+	Rank 1	3/4	Strong	no FDR	Transport
<i>CLDN14</i>	+	Rank 1	3/4	Strong	no FDR	Ion
<i>CASR</i>	+	Rank 1	3/4	Strong	no FDR	Ca sense
<i>HIBADH</i>	+	Rank 1	3/4	Strong	no FDR	Support
<i>PKD2</i>	+	Rank 1	3/4	Broad	nominal	Broad epi.

Bulk: no uniform FDR-supported P1 single-gene response